Optimal Time Reuse in Cooperative D2D Relaying Networks

Zhaowei Zhu, Shengda Jin and Xiliang Luo
School of Information Science and Technology, ShanghaiTech University,
Shanghai, China
Email: luoxl@shanghaitech.edu.cn

***Abstract*—Device-to-device (D2D) communication has become one important part of the 5G cellular networks particularly due to the booming of proximity-based applications, e.g. D2D relays. However, the D2D relays may create strong interference to nearby users that are sharing the same spectrum. Thus interference management in a cooperative network with D2D relays is critical. In this paper, we study the optimal time reuse patterns in a cellular network with cooperative D2D relays and derive the corresponding optimized relaying strategies. Due to the binary association constraints and the fact that the total number of feasible time reuse patterns increase exponentially with the number of servers, we have a large scale integer programming problem which is formidable to solve. However, we show that we just need to activate a few time reuse patterns in the order of the number of users. Accordingly, a low-complexity algorithm is proposed to find the set of active time reuse patterns and solve this problem in an approximated way. Numerical simulations demonstrate that our scheme can efficiently allocate the time resources and determine the relaying strategies. Furthermore, the proposed scheme offers significant gains relative to the existing ones.**

***Index Terms*—Cooperative network, device-to-device relay, time reuse, resource allocation, spectrum sharing.**

I. Introduction

With the ever-increasing mobile data traffic demands, the proximity-based communication through the device-to-device (D2D) links is regarded as one promising technology in 5G cellular networks [1]. Different from the base station (BS)-based links, D2D links usually appear in short-range scenarios and can significantly enhance the network throughput performance without interfering distant users [2]. Moreover, due to the scarcity of licensed spectrum for conventional cellular networks, increasing attentions have been paid to the efficient utilization of all possible spectrum resources including the unlicensed spectrum to offer ubiquitous and seamless access to mobile users [3]. When both cellular networks and D2D networks share the same unlicensed spectrum resources as described in [3], severe interference can still occur among nearby users without careful resource management [4].

To make full use of the limited spectrum resources, a lot of works have been done to optimize D2D-enabled cooperative networks [5]–[9]. In particular, the spectrum allocation and power control were jointly optimized in [5] to maximize different network utility metrics under QoS constraints, while D2D relays were not modeled in [5]. Recent works in [6]–[9] emphasized the importance of D2D relays and recommended some D2D user equipments (DUEs) to serve as D2D relays for other DUEs experiencing poor BS-based links. Orthogonal resource allocations and corresponding relaying strategies were optimized jointly under different constraints in [6]–[9]. Specifically, the orthogonal time/frequency resources were allocated between the BS and the D2D relays in [6], among the neighboring nodes in [7], and among the D2D relays in [8], [9]. However, we know non-orthogonal resource reuse among the BSs and the D2D relays will further enhance the network throughput performance. Each reuse strategy was modeled as one reuse pattern and the frequency resource allocations were optimized over all possible reuse patterns for heterogeneous networks (HetNets) in [10]–[12]. In particular, the network-wide average packet delay was minimized in [10], the energy efficiency was maximized in [11], and the network proportional fairness (PF) metric was maximized in [12]. Nowadays, ubiquitous DUEs are playing an increasingly important role in offloading data traffic from the heavily loaded cellular network, thus the optimal reuse scheme for a cooperative D2D network is also urgent to be studied for making full use of limited resources.

Different from those previous works, we endeavor to find the optimal resource reuse scheme in a cooperative network with D2D relays. The major contributions can be summarized as follows. Firstly, when both cellular networks and D2D networks utilize the same licensed/unlicensed spectrum resources, we study the optimal set of time reuse patterns and the corresponding relaying strategies such that the PF metric of the network is maximized. Even though the total number of feasible reuse patterns increases exponentially with the number of users, we show that we just need to activate a few time reuse patterns in the order of the number of users to achieve the optimal network performance. Secondly, we put forward an efficient algorithm of low complexity to identify the set of reuse patterns to be activated and solve the problem approximately. Numerical results demonstrate our scheme can offer significant gains relative to the existing solutions.

The remainder of this paper is organized as follows. Section II describes the system model. Section III formulates the optimization problem and characterizes the optimal time reuse profile. Section IV proposes one low-complexity algorithm. Numerical results are given in Section V and Section VI concludes the paper.

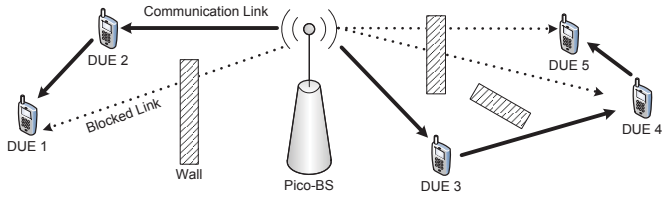


Fig. 1. A cooperative network with D2D relays. DUE 1 is served by DUE 2 since its BS-based link is blocked. Both DUE 4 and DUE 5 are served by DUE 3 for similar reasons.

Notations: Notations $\|\boldsymbol{x}\|_0$, $\boldsymbol{A}^T$, $\mathsf{conv}(\mathcal{A})$, $|\mathcal{A}|$, and $\boldsymbol{a} \oplus \boldsymbol{b}$ stand for l_0-norm of vector $\boldsymbol{x}$, the transpose of matrix $\boldsymbol{A}$, the convex hull of the set $\mathcal{A}$, the cardinality of the set $\mathcal{A}$, and the element-wise XOR operation of two vectors. $\mathbb{1}\{\cdot\}$ stands for the indicator function and takes the value of 1 (0) when the specified condition is met (otherwise). $\boldsymbol{e}_n$ denotes the unit vector with only the n-th element being 1 and all other elements being 0.

II. System Model

As shown in Fig. 1, a downlink shared cooperative network [7] is considered here, where a number of nodes coexist including the DUEs and the BSs. The sets of BSs and DUEs are denoted by $\mathcal{B}$ and $\mathcal{U}$, respectively. Meanwhile, we define $B := |\mathcal{B}|$ and $U := |\mathcal{U}|$. Each DUE is assumed to be willing to act as either a relay or a user. For instance, it can act as a user to receive data either from the serving BS or from another DUE (behave as one D2D relay) in one time slot. In another slot, it can act as a relay to help other DUEs. Thus we assume each DUE can choose to be either a user or a relay in different time slots. The cost aversion of helping others may vary from DUEs and can be left for future works, e.g. energy or data plan cost aversion. The set of servers $\mathcal{N}$ including both all BSs and D2D relays is defined as follows.

$$\mathcal{N} := \{\underbrace{1, \cdots, B}_{\text{BSs}}, \underbrace{B+1, \cdots, B+U}_{\text{potential DUEs}}\}. \tag{1}$$

Let $N := |\mathcal{N}|$ denote the total number of servers in the set $\mathcal{N}$. We use a vector $\boldsymbol{v}_i = [v_{i,1}, \cdots, v_{i,N}]^T$ to indicate the active status of the servers in one time slot and call it the i-th time reuse pattern. The n-th element of $\boldsymbol{v}_i$ is 1, i.e. $v_{i,n} = 1$, when server n is active under time reuse pattern i. Otherwise, we have $v_{i,n} = 0$. Note when a server is active, we assume it utilizes the whole downlink bandwidth for information transmission. There are totally $2^N - 1$ feasible time reuse patterns to be considered and we do not consider the all 0 pattern, which means nobody transmits.

Let $\mathcal{I} = \{1, \cdots, 2^N - 1\}$ denote the set of indices of all the feasible time reuse patterns. Furthermore, the set of all time reuse pattern is denoted as $\tilde{\mathcal{V}} = \{\boldsymbol{v}_i | i \in \mathcal{I}\}$. We also use $\mathcal{A}_i$ to denote the set of active servers under pattern i, i.e.

$$\mathcal{A}_i := \{n | v_{i,n} = 1, n \in \mathcal{N}\}. \tag{2}$$

Let x_i denote the fraction of the total time duration of length T allocated to pattern i. The overall time reuse profile is thus given by the vector: $\boldsymbol{x} := [x_1, \cdots, x_{2^N-1}]^T$. In order to get the optimal network performance, we need to utilize the whole time duration, i.e.

$$\sum_{i \in \mathcal{I}} x_i = 1. \tag{3}$$

Furthermore, we assume time-division multiple-access (TDMA) and each server only transmits data to one served user at one time. A server needs to divide the allocated time resource under pattern i orthogonally among its served users. We use $y_{u,n,i} \geq 0$ to denote the fraction of the total duration T allocated to user u by server n under reuse pattern i. Clearly, it is bounded as

$$\sum_{u \in \mathcal{U}} y_{u,n,i} \leq x_i. \tag{4}$$

Additionally, as in the current cellular networks, each user is allowed to be associated to a single server under each time reuse pattern, which is characterized by $z_{u,n,i}$, i.e. $z_{u,n,i} = 1$ indicates user u is served by server n under pattern i and $z_{u,n,i} = 0$ otherwise. Due to this single-server association constraint, the variables $\{z_{u,n,i}\}$ satisfy the constraint $\sum_{n \in \mathcal{N}} z_{u,n,i} = 1$. Note the corresponding relaying strategies are also implied by $\{z_{u,n,i}\}$.

Denote the transmission power of server n by P_n. The signal-to-interference-plus-noise ratio (SINR) of the link between user u and server n under pattern i is denoted by $\gamma_{u,n,i}$ and can be derived as

$$\gamma_{u,n,i} = \frac{\mathbb{1}\{n \in \mathcal{A}_i\} P_n |g_{n,u}|^2}{\sigma^2 + \sum_{m \in \mathcal{A}_i, m \neq n} P_m |g_{m,u}|^2}, \tag{5}$$

where σ^2 is the power of the thermal noise at the user and $g_{n,u}$ denotes the average gain of the channel from server n to user u. Let $c_{u,n,i}$ denote the spectral efficiency of this link. To reflect the constraint that one DUE cannot transmit and receive simultaneously, the spectral efficiency $c_{u,n,i}$ is set to be zero if server $(u + B)$ is active under reuse pattern i. Therefore, the spectral efficiency can be expressed as

$$c_{u,n,i} = \mathbb{1}\{(u + B) \notin \mathcal{A}_i\} \cdot \log_2(1 + \gamma_{u,n,i}). \tag{6}$$

Note that the actual time resource allocated to user u by server n under pattern i is $T y_{u,n,i}$. The average data rate of user u over all reuse patterns can be written as follows.

$$\begin{aligned} \bar{R}_u &= \frac{1}{T} \cdot W \cdot \sum_{i \in \mathcal{I}} \sum_{n \in \mathcal{N}} z_{u,n,i} \cdot T \cdot y_{u,n,i} \cdot c_{u,n,i} \\ &= W \cdot \sum_{i \in \mathcal{I}} \sum_{n \in \mathcal{N}} z_{u,n,i} \cdot y_{u,n,i} \cdot c_{u,n,i}, \end{aligned} \tag{7}$$

where W denotes the system bandwidth. Besides, we assume that the DUEs in the network demand different data. For a D2D relay, a portion of its received data are intended for others. The effective data rate of user u is thus given by $R_u = \bar{R}_u - \tilde{R}_{(u+B)}$, where $\tilde{R}_{(u+B)}$ denotes the service data rate of server $(u + B)$, i.e. user u acting as a D2D relay, targeted to the served DUEs. The total rate of server n serving other DUEs over all reuse patterns is given by

$$\tilde{R}_n = W \sum_{i \in \mathcal{I}} \sum_{u \in \mathcal{U}} z_{u,n,i} \cdot y_{u,n,i} \cdot c_{u,n,i}. \tag{8}$$

The single-server association rule results in a combinatorial problem and finding an optimal solution need the exhaustive search. In the next section, we will relax the single-server constraint and allow one user to be served by multiple servers. This will enable low-complexity algorithms to find the optimal time reuse profile.

III. Number of Active Time Reuse Patterns

$$\underset{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}}{\text{maximize}} \quad \sum_{u\in\mathcal{U}} \log(R_u) \tag{9a}$$

$$\text{subject to} \quad \bar{R}_u = W \sum_{i\in\mathcal{I}} \sum_{n\in\mathcal{N}} z_{u,n,i} \cdot y_{u,n,i} \cdot c_{u,n,i}, \forall u \tag{9b}$$

$$\tilde{R}_n = W \sum_{i\in\mathcal{I}} \sum_{u\in\mathcal{U}} z_{u,n,i} \cdot y_{u,n,i} \cdot c_{u,n,i}, \forall n \tag{9c}$$

$$R_u = \bar{R}_u - \tilde{R}_{u+B}, \ R_u > 0, \forall u \tag{9d}$$

$$\sum_{n\in\mathcal{N}} z_{u,n,i} = 1, z_{u,n,i} \in \{0,1\}, \forall u,n,i \tag{9e}$$

$$x_i \geq \sum_{u\in\mathcal{U}} z_{u,n,i} \cdot y_{u,n,i}, \forall n,i \tag{9f}$$

$$\sum_{i\in\mathcal{I}} x_i = 1, x_i \geq 0, \forall i, \ y_{u,n,i} \geq 0, \forall u,n,i \tag{9g}$$

In this section, we would like to maximize the PF metric of the network through optimizing the time reuse profile. To this end, we formulate the optimization problem (9), where the optimization variables $\boldsymbol{x}$, $\boldsymbol{y}$, $\boldsymbol{z}$ are the overall time reuse profile: $\{x_i, \forall i\}$, the overall user time allocation profile: $\{y_{u,n,i}, \forall u,n,i\}$, and the DUE association profile: $\{z_{u,n,i}, \forall u,n,i\}$ respectively. Problem (9) is a large knapsack problem, which is NP-hard to find its optimal solution.

By allowing fractional user association, we can drop the single-server association indicator variable $\boldsymbol{z}$ and rely on $\boldsymbol{y}$ to imply the fractional user association profile. Particularly, given $y_{u,n,i} > 0$, we can infer the u-th user is served by the n-th server under pattern i. The relaxed multi-server association problem is shown in (10), whose solution offers an upper bound of the problem in (9). Since the active servers in each pattern i will use all the available time resources to optimize the network performance, the inequalities in (9f) indeed become the equalities in (10e).

The objective function in (10) is concave and all the constraints are linear. However, the amount of variables, i.e. $\{x_i\}$ and $\{y_{k,n,i}\}$, increases exponentially with the number of servers due to the fact that we have $2^N - 1$ different time reuse patterns. For a large cooperative network, it becomes a formidable task to find the optimal solution to the convex problem in (10). Fortunately, we can show that only a very limited number of reuse patterns need to be activated without sacrificing the network throughput performance.

Proposition 1. *There exists one optimal solution to the problem in (10) where at most U out of $2^N - 1$ reuse patterns are active. Specifically, we can find one time reuse profile $\boldsymbol{x}^*$ which maximizes the objective function in (10) and satisfies the following criterion:*

$$\underset{\boldsymbol{x},\boldsymbol{y}}{\text{maximize}} \quad \sum_{u\in\mathcal{U}} \log(R_u) \tag{10a}$$

$$\text{subject to} \quad \bar{R}_u = \sum_{i\in\mathcal{I}} \sum_{n\in\mathcal{N}} y_{u,n,i} \cdot c_{u,n,i}, \forall u \tag{10b}$$

$$\tilde{R}_n = \sum_{i\in\mathcal{I}} \sum_{u\in\mathcal{U}} y_{u,n,i} \cdot c_{u,n,i}, \forall n \tag{10c}$$

$$R_u = \bar{R}_u - \tilde{R}_{u+B}, \ R_u > 0, \forall u \tag{10d}$$

$$x_i = \sum_{u\in\mathcal{U}} y_{u,n,i}, \forall n,i \tag{10e}$$

$$\sum_{i\in\mathcal{I}} x_i = 1, x_i \geq 0, \forall i, \ y_{u,n,i} \geq 0, \forall u,n,i \tag{10f}$$

$$\|\boldsymbol{x}^*\|_0 \leq U. \tag{11}$$

Proof. Suppose we are given one particular optimal solution to the problem in (10), i.e. $\boldsymbol{x}$ and $\boldsymbol{y}$. Each element of $\boldsymbol{y}$ can be re-written as $y_{u,n,i} = x_i \cdot \tau_{u,n,i}$. The rates $\bar{R}_u$ and $\tilde{R}_n$ in (10b) and (10c) can be expressed as

$$\bar{R}_u = W \sum_{i\in\mathcal{I}} x_i \bar{R}_u^i, \ \tilde{R}_n = W \sum_{i\in\mathcal{I}} x_i \tilde{R}_n^i, \tag{12}$$

where

$$\bar{R}_u^i := \sum_{n\in\mathcal{N}} \tau_{u,n,i} \cdot c_{u,n,i} \tag{13}$$

and

$$\tilde{R}_n^i := \sum_{u\in\mathcal{U}} \tau_{u,n,i} \cdot c_{u,n,i}. \tag{14}$$

To describe the property of the effective data rate vector $\boldsymbol{R} := [R_1, \cdots, R_U]^T$ clearly, we define the receiving and transmitting rate vectors as

$$\bar{\boldsymbol{R}} := [\bar{R}_1, \cdots, \bar{R}_U]^T, \ \tilde{\boldsymbol{R}} := [\tilde{R}_{B+1}, \cdots, \tilde{R}_N]^T. \tag{15}$$

Accordingly, the receiving and transmitting rate vectors under time reuse pattern i is defined as

$$\bar{\boldsymbol{R}}^i := [\bar{R}_1^i, \cdots, \bar{R}_U^i]^T, \ \tilde{\boldsymbol{R}}^i := [\tilde{R}_{B+1}^i, \cdots, \tilde{R}_N^i]^T. \tag{16}$$

Combining (10d), (12), and the definitions of corresponding vectors, we have

$$\boldsymbol{R} = \left[\boldsymbol{\Phi}^1, \cdots, \boldsymbol{\Phi}^{2^N-1}\right] \boldsymbol{x}, \tag{17}$$

where $\boldsymbol{\Phi}^i := \bar{\boldsymbol{R}}^i - \tilde{\boldsymbol{R}}^i$. By defining a set $\mathcal{Q}$ as

$$\mathcal{Q} := \{\boldsymbol{\Phi}^1, \boldsymbol{\Phi}^2, ..., \boldsymbol{\Phi}^{2^N-1}\}, \tag{18}$$

Thus R lies in conv(Q). Let p be the affine dimension of conv(Q), so p <= U. Caratheodory gives a representation using at most p+1 points. If p < U, this is already at most U active patterns. It remains to consider the full-dimensional case p = U.

When p = U, the sum-log objective is strictly increasing on the positive effective-rate domain. An optimum is therefore on a supporting Pareto face of conv(Q), not in its full-dimensional interior. That exposed face has affine dimension at most U-1, so Caratheodory applied on the face represents R* with at most U points. Together with the p < U case, there exists an optimal profile x* with ||x*||_0 <= U. This argument uses the effective rate vector only; it does not require every intermediate link rate to be positive.

$$\|\boldsymbol{x}^*\|_0 \leq U, \quad \boldsymbol{R}^* = [\boldsymbol{\Phi}^1, ..., \boldsymbol{\Phi}^{2^N-1}]\boldsymbol{x}^*. \tag{19}$$

□

Proposition 1 indicates that we only need to turn on no more than U reuse patterns to achieve the optimal network throughput performance instead of activating all the $2^N - 1$ feasible patterns.

IV. Efficient Pattern Selection and Resource Allocation Algorithm

As indicated by Proposition 1, the following optimization problem shares the same optimal objective value as the problem in (10).

$$\underset{\boldsymbol{x},\boldsymbol{y},\mathcal{V}}{\text{maximize}} \quad P(\boldsymbol{x},\boldsymbol{y},\mathcal{V}) = \sum_{u\in\mathcal{U}} \log(R_u) \tag{20a}$$

$$\text{subject to} \quad x_i = 0, i \in \{i|\boldsymbol{v}_i \notin \mathcal{V}\}, (10\text{b}) - (10\text{f}), \tag{20b}$$

$$|\mathcal{V}| \leq U, \tag{20c}$$

where $\mathcal{V} \subseteq \tilde{\mathcal{V}}$ denotes the candidate set of reuse patterns containing at most U patterns. The problem in (20) is still very hard to solve. We split it into two subproblems and solve them iteratively. In particular, during the $(t+1)$-th iteration, we carry out the following updates.

• *P1. Pattern Selection*: Determine the pattern set $\mathcal{V}^{t+1}$ such that

$$\mathcal{V}^{t+1} \subseteq \tilde{\mathcal{V}}, \ |\mathcal{V}^{t+1}| \leq U; \tag{21}$$

• *P2. Resource Allocation:* Update the resource allocation and the corresponding association rules as

$$\underset{\boldsymbol{x}\in\mathcal{G}(\mathcal{V}^{t+1})}{\text{maximize}} \ \underset{\boldsymbol{y}\in\mathcal{F}(\boldsymbol{x})}{\text{maximize}} \ P(\boldsymbol{x},\boldsymbol{y},\mathcal{V}^{t+1}), \text{ subject to } (20\text{b}), \tag{22}$$

where

$$\mathcal{G}(\mathcal{V}) := \{\boldsymbol{x}| \sum_{i\in\{j|\boldsymbol{v}_j\in\mathcal{V}\}} x_i = 1, x_i \geq 0, \forall i\} \tag{23}$$

and

$$\mathcal{F}(\boldsymbol{x}) := \{\boldsymbol{y}| \sum_u y_{u,n,i} = x_i, y_{u,n,i} \geq 0, \forall u, n, i\}. \tag{24}$$

The problem in (22) is convex and can be solved efficiently by general convex solvers, e.g. CVX [15]. However, the pattern selection subproblem has $\sum_{m=1}^{U} \binom{2^N-1}{m}$ possible solutions in theory and it is impractical to apply the exhaustive search method. Next, we propose an approximated solution enjoying low-complexity.

From $\mathcal{V}^t$, we can define a set $\mathcal{V}_d^t$ as

$$\mathcal{V}_d^t := \{\boldsymbol{v}\big|\boldsymbol{v} \in \tilde{\mathcal{V}}, \exists \boldsymbol{v}' \in \mathcal{V}^t, D_H(\boldsymbol{v},\boldsymbol{v}') \leq d\}, \tag{25}$$

where $d \leq N$ and $D_H(\boldsymbol{v},\boldsymbol{v}')$ denotes the Hamming distance between patterns $\boldsymbol{v}$ and $\boldsymbol{v}'$. Furthermore, we add a particular pattern set to the candidate pattern set $\mathcal{V}^t$ in the $(t+1)$-th iteration when $|\mathcal{V}^t| < U$, i.e. $\mathcal{V}^{t+1} = \mathcal{V}^t \bigcup \mathcal{V}^*$ and $\mathcal{V}^* \subseteq \mathcal{V}_d^t$ denotes the pattern set to be included in the $(t+1)$-th iteration. Inspired by the pattern selection approach in [12] and the Frank-Wolfe method [13], we put forward an iterative pattern selection method according to the solution of the following problem:

$$(\mathcal{V}^*, \boldsymbol{x}^*, \boldsymbol{y}^*) = \arg \max_{\mathcal{V}\subseteq\mathcal{V}_d^t} \max_{\boldsymbol{x}\in\mathcal{G}(\mathcal{V})} \max_{\boldsymbol{y}\in\mathcal{F}(\boldsymbol{x})} \nabla_{\boldsymbol{y}} P(\boldsymbol{x}^t, \boldsymbol{y}^t, \mathcal{V}^t)^T \boldsymbol{y}. \tag{26}$$

Basically, in (26), we identify a set of reuse patterns to activate by finding the direction providing the most predominant improvement in the objective value under the specified constraints. Denoting the (u,n,i)-th entry of the gradient vector $[\nabla_{\boldsymbol{y}} P(\boldsymbol{x}^t, \boldsymbol{y}^t, \mathcal{V}^t)]_{u,n,i}$ by $p_{u,n,i}^t$, we can have

$$p_{u,n,i}^t = W \cdot c_{u,n,i} \cdot \left(R_u^{-1} - \mathbb{1}\{n > B\} \cdot R_{n-B}^{-1}\right). \tag{27}$$

Now we can establish the following result.

Proposition 2. *Among the solution to the problem in (26), the set $\mathcal{V}^*$ contains the following time reuse pattern:*

$$\boldsymbol{v}_{i^*} = \arg \max_{\boldsymbol{v}_i\in\mathcal{V}_d^t} \sum_{n\in\mathcal{N}} p_{u_{n,i}^*,n,i}^t. \tag{28}$$

where $u_{n,i}^* := \arg\max_u \ p_{u,n,i}^t$.

Proof. Defining $\tau_{u,n,i} := y_{u,n,i}/x_i$, the problem in (26) can be re-written as

$$\max_{\mathcal{V}\subseteq\mathcal{V}_d^t} \max_{\boldsymbol{x}\in\mathcal{G}(\mathcal{V})} \sum_{i\in\mathcal{I}} x_i \sum_{n\in\mathcal{N}} \max_{\sum_u \tau_{u,n,i}=1, \atop \tau_{u,n,i}\geq 0} \sum_{u\in\mathcal{U}} \tau_{u,n,i} \cdot p_{u,n,i}^t. \tag{29}$$

The third maximization is a constrained linear programming (LP) problem and the optimal $\tau_{u,n,i}^*$ is given by

$$\tau_{u,n,i}^* = \mathbb{1}\{u = u_{n,i}^*\}, \text{where } u_{n,i}^* := \arg\max_u p_{u,n,i}^t. \tag{30}$$

The problem in (29) thus becomes

$$\max_{\mathcal{V}\subseteq\mathcal{V}_d^t} \max_{\boldsymbol{x}\in\mathcal{G}(\mathcal{V})} \sum_{i\in\mathcal{I}} x_i \sum_{n\in\mathcal{N}} p_{u_{n,i}^*,n,i}^t. \tag{31}$$

The problem in (31) is again a constrained LP and the solution is: $x_i^* = \mathbb{1}\{\boldsymbol{v}_i = \boldsymbol{v}_{i^*}\}$, $\boldsymbol{v}_{i^*} = \arg\max_{\boldsymbol{v}_i\in\mathcal{V}_d^t} \sum_{n\in\mathcal{N}} p_{u_{n,i}^*,n,i}^t$. □

The solution to (26) during the $(t+1)$-th iteration simply tells us to activate the particular time reuse pattern $\boldsymbol{v}_{i^*}$ to maximize the improvement in the objective value. Clearly, the associated complexity in finding the candidate pattern set $\mathcal{V}^{t+1}$ is determined by the size of $\mathcal{V}_d^t$. We have the lowest complexity in the pattern selection subproblem by setting $d = 1$. To enable

Algorithm 1 Optimal time reuse with D2D relays.

1: **Initialization:** $\epsilon_1 = \epsilon_2 = 10^{-4}$, $t = 1$, $\mathcal{V}^1 = \{\boldsymbol{e}_1 + \boldsymbol{e}_{u+1} | u \in \mathcal{U}\}$, and $P^0 = -1$. Solve (22) with $\mathcal{V}^1$, obtain $\boldsymbol{x}^1$, $\boldsymbol{y}^1$, and P^1;
2: **While** $|P^t - P^{t-1}| > \epsilon_2$ and $|\mathcal{V}^t| \leq U$
3: Choose the active time reuse patterns $\{\boldsymbol{v}_{i_n^*}, n \in \mathcal{N}\}$ as (32)-(33);
4: Obtain $\hat{\mathcal{V}}^{t+1}$ with $\{\boldsymbol{v}_{i_n^*}\}_{n\in\mathcal{N}}$ as (34);
5: Solve (22) with $\hat{\mathcal{V}}^{t+1}$, obtain $\boldsymbol{x}^{t+1}$, $\boldsymbol{y}^{t+1}$, and P^{t+1};
6: Compress to at most U patterns as in (35);
7: $t = t + 1$;
8: **End**
9: Determine the single-server associations as (36);
10: Solve (9) with $\mathcal{V}^t$ and the derived associations to get the optimal resource allocations and corresponding relaying strategies.

faster convergence, we select N time reuse patterns in each iteration. Specifically, we modify the ON/OFF state of the n-th server in each reuse pattern $\boldsymbol{v}_i \in \mathcal{V}^t$ as $\boldsymbol{v}_{i_n} := \boldsymbol{v}_i \oplus \boldsymbol{e}_n$ and define the set $\bar{\mathcal{V}}_n^t$ as

$$\bar{\mathcal{V}}_n^t = \{\boldsymbol{v}_{i_n} | \boldsymbol{v}_{i_n} := \boldsymbol{v}_i \oplus \boldsymbol{e}_n, \boldsymbol{v}_i \in \mathcal{V}^t, \boldsymbol{v}_i \neq \boldsymbol{e}_n\}. \quad (32)$$

Then we determine a new candidate reuse pattern $\boldsymbol{v}_{i_n^*}$ to activate according to Proposition 2 as

$$\boldsymbol{v}_{i_n^*} = \arg \max_{\boldsymbol{v}_{i_n} \in \bar{\mathcal{V}}_n^t, \boldsymbol{v}_{i_n} \notin \mathcal{V}^t} \sum_{\bar{n}\in\mathcal{N}} p^t_{u^*_{\bar{n},i},\bar{n},i}. \quad (33)$$

Now a temporary reuse pattern set for the $(t+1)$-th iteration, i.e. $\hat{\mathcal{V}}^{t+1}$ can be derived as

$$\hat{\mathcal{V}}^{t+1} = \mathcal{V}^t \cup \{\boldsymbol{v}_{i_n^*}, n \in \mathcal{N}\}. \quad (34)$$

The union in (34) is temporary and may exceed U patterns. Solve (22) on that union. At this optimum, apply Proposition 1 on its supporting face: choose nonnegative weights on at most U patterns that preserve the effective rate vector, hence the sum-log objective. Adopt that compressed allocation and its support as V^(t+1) before the next iteration. Threshold trimming alone does not guarantee the budget. Thus |V^(t+1)| <= U. (35)

Note that the solution obtained by solving (20) allows multi-server association. To meet the single-server association requirement in (9), we simply associate the user to the server that gives the largest data rate under the particular time reuse pattern $\boldsymbol{v}_i$, i.e.

$$z_{u,n,i} = \mathbb{1}\{n = n^*_{u,i}\},\ n^*_{u,i} = \arg\max_{n\in\mathcal{N}} y_{u,n,i} \cdot c_{u,n,i}. \quad (36)$$

After finalizing the user association rule as in (36), the original problem in (9) can be solved with low complexity since $\boldsymbol{z}$ is given and the number of active patterns in $\mathcal{V}^t$ is limited. Algorithm 1 summarizes all the steps.

TABLE I
BRUTE-FORCE CVX VS. PROPOSED LOW-COMPLEXITY SOLUTION.

# DUEs	10	11	12	···	30
(θ_B, θ_L)	(8.63, 8.63)	(8.61, 8.51)	(7.12, 7.11)	···	(?, 7.34)
(t_1, t_2)	(735, 20)	(3004, 40)	(10608, 45)	···	(?, 160)

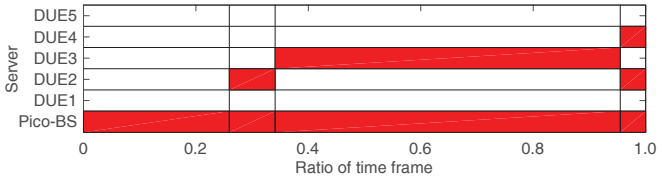


Fig. 2. The active time reuse patterns obtained from the proposed algorithm with respect to the network in Fig. 1. Solid color indicates that the particular time slot is occupied by the corresponding server.

V. NUMERICAL RESULTS

In this section, we test our algorithm by simulating a cooperative network with only one pico-BS (30dBm transmission power (TxPwr)) since we focus on the relay behavior of the DUEs (20dBm TxPwr). See also Fig. 1 for one example. Other parameters are set as follows.

- DUEs are uniformly dropped in a square specified by $[0, 200]$m$\times[0, 200]$m and the pico-BS is deployed at the center;
- System bandwidth is 20MHz. Noise PSD is -174dBm/Hz. The path-loss is determined as $37.6 \log_{10} d_m + 35.3 + 5n_w$(dB) [5], where d_m is the distance in meters, and n_w stands for the number of walls [16].

Firstly, we compare the geometric mean (GM) of the DUEs' throughputs and the CPU running time between the brute-force optimal solution (solve the problem in (10) directly with CVX) and our proposed low-complexity solution in Table I. In this table, θ_B denotes the GM throughput in Mbps with the brute-force solution and θ_L denotes the one with our proposed algorithm. Additionally, t_1 and t_2 are the consumed CPU time of the brute-force solution and our proposed algorithm in seconds. Note that there is a the question mark "?" in the case of 30 DUEs, which indicates that the brute-force CVX solution can not be obtained with our lab computer. From the table, we can find out that our proposed algorithm performs satisfactorily and only suffers a very small amount of degradation as we have 12 DUEs.

Secondly, we would like to show that the optimal resource reuse profile is non-orthogonal. We simulate a cooperative network with 5 DUEs as illustrated in Fig. 1, where some DUEs suffer poor channel conditions due to the walls. The set of activated time reuse patterns and the fraction of total time allocated to each pattern are shown in Fig. 2. The total number of active time reuse patterns is 4, which follows Proposition 1. In this figure, we find that the results are consistent with the intuitive solution. For example, DUE 3 is active for most of time such that it can relay enough data to DUE 4 and DUE 5. In the last time slot, both DUE 2 and DUE 4 behave as D2D relays to serve DUE 1 and DUE 5, whose links to the BS are blocked by the walls. It also indicates that the optimal time

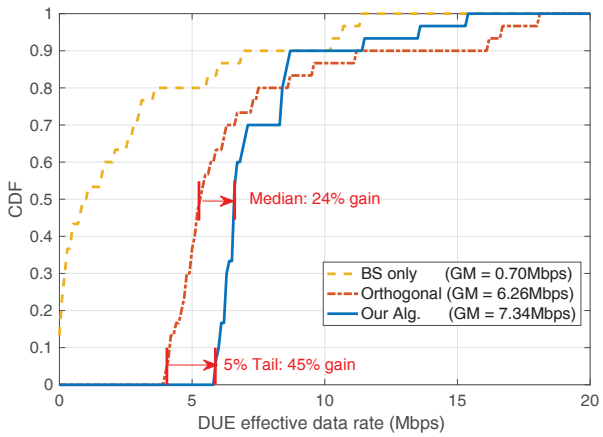


Fig. 3. The CDF of 30 DUE effective data rates. Our Alg.: solution with the proposed algorithm. Orthogonal: solution with the orthogonal scheme [9]. BS only: solution without D2D relays.

reuse patterns are not as those proposed in [6]–[9].

Furthermore, we compare the effectiveness of our proposed time reuse scheme with some existing schemes including the orthogonal scheme and the BS only scheme. The orthogonal scheme in Fig. 3 considers the time reuse patterns where only the BS and one D2D relay are active in one time slot [9], i.e. $\mathcal{V} = \{\boldsymbol{e}_1 + \boldsymbol{e}_{u+1} | u \in \mathcal{U}\}$. In Fig. 3, some interesting observations are worthy to be pointed out. On the one hand, the minimum effective data rate obtained with our algorithm is about 5.8Mbps, which is greater than almost 60% and 80% of the DUEs with the orthogonal scheme and the BS only scheme respectively. On the other hand, the orthogonal scheme results in more high rate DUEs than our scheme. For example, 13% of the DUEs enjoy effective data rates more than 10Mbps in the orthogonal case while only 10% of the DUEs get similar rates in the optimal reuse case. Our proposed algorithm will ask those DUEs with high data rates to serve as D2D relays to help the DUEs with poor channel conditions. It is clear that our proposed algorithm offers significant median and tail gains relative to the orthogonal scheme. Moreover, our proposed algorithm achieves higher GM data rate than the other two existing schemes.

VI. Conclusions

In this paper, we have studied the optimal time reuse patterns and the corresponding relaying strategies in a cooperative network with D2D relays. The original optimization problem is of a formidable size. This is due to the fact that the total number of feasible time reuse patterns scales exponentially with the number of nodes in the network. To circumvent this dilemma, we have shown that we just need to turn on a limited number of time reuse patterns without sacrificing the network performance firstly. In particular, we have proved that the number of active reuse patterns can be no more than the number of users. Secondly, we have put forward a low-complexity algorithm to identify the small set of active time reuse patterns and solve the large scale optimization problem approximately. Compared to those existing schemes with orthogonal resource allocation constraints, our proposed scheme offers significant gains.

References

[1] M. N. Tehrani, M. Uysal, and H. Yanikomeroglu, "Device-to-device communication in 5G cellular networks: Challenges, solutions, and future directions," *IEEE Commun. Mag.*, vol. 52, no. 5, pp. 86–92, May 2014.

[2] X. Xu, Y. Zhang, Z. Sun, Y. Hong, and X. Tao, "Analytical modeling of mode selection for moving D2D-enabled celluar networks," *IEEE Commun. Lett.*, vol. 20, no. 6, pp. 1203–1206, Jun. 2016.

[3] H. Zhang, X. Chu, W. Guo and S. Wang, "Coexistence of Wi-Fi and heterogeneous small cell networks sharing unlicensed spectrum," *IEEE Commun. Mag.*, vol. 53, no. 3, pp. 158–164, Mar. 2015.

[4] Y. Liu, R. Wang, and Z. Han, "Interference-constrained pricing for D2D networks," *IEEE Trans. Wireless Commun.*, vol. 16, no. 1, pp. 475–486, Jan. 2017.

[5] X. Ma, J. Liu, and H. Jiang, "Resource allocation for heterogeneous applications with device-to-device communication underlaying cellular networks," *IEEE J. Sel. Areas Commun.,* vol. 34, no. 1, pp. 15–26, Jan. 2016.

[6] B. Zhou, H. Hu, S. Q. Huang, and H. H. Chen, "Intracluster device-to-device relay algorithm with optimal resource utilization," *IEEE Trans. Veh. Technol.*, vol. 62, no. 5, pp. 2315–2326, Jun. 2013.

[7] M. Li, S. Salinas, P. Li, X. Huang, Y. Fang, and S. Glisic, "Optimal scheduling for multi-radio multi-channel multi-hop cognitive cellular networks," *IEEE Trans. Mobile Comput.*, vol. 14, no. 1, pp. 139–154, Jan. 1 2015.

[8] H. Xu and B. Li, "Resource allocation with flexible channel cooperation in cognitive radio networks," *IEEE Trans. Mobile Comput.*, vol. 12, no. 5, pp. 957–970, May 2013.

[9] N. Mastronarde, V. Patel, J. Xu, L. Liu, and M. van der Schaar, "To relay or not to relay: learning device-to-device relaying strategies in cellular networks," *IEEE Trans. Mobile Comput.*, vol. 15, no. 6, pp. 1569–1585, Jun. 2016.

[10] B. Zhuang, D. Guo, and M. L. Honig, "Traffic-driven spectrum allocation in heterogeneous cellular networks," *IEEE J. Sel. Areas Commun.*, vol. 33, no. 10, pp. 2027–2038, Oct. 2015.

[11] B. Zhuang, D. Guo, and M. L. Honig, "Energy-efficient cell activation, user association, and spectrum allocation in heterogeneous networks," *IEEE J. Sel. Areas Commun.*, vol. 34, no. 4, pp. 823–831, Apr. 2016.

[12] Q. Kuang, W. Utschick, and A. Dotzler, "Optimal joint user association and multi-pattern resource allocation in heterogeneous networks," *IEEE Trans. Signal Process.*, vol. 64, no. 13, pp. 3388–3401, Jul. 2016.

[13] M. S. Bazaraa, H. D. Sherali, and C. M. Shetty, *Nonlinear Programming: Theory and Algorithms*, 3rd ed. Hoboken, NJ, USA: Wiley, 2006.

[14] D. Bertsimas and J. N. Tsitsiklis, *Introduction to Linear Optimization*. Belmont, MA, USA: Athena Scientific, 1997.

[15] M. Grant and S. Boyd. (Sep. 2013). *CVX: MATLAB Software for Disciplined Convex Programming, Version 2.0 Beta*. [Online]. Available: http://cvxr.com/cvx

[16] J. Meinilä, P. Kyösti, T. Jämsä, and L. Hentilä, "WINNER II channel models," in *Radio Technologies and Concepts for IMT-Advanced. Hoboken*, NJ, USA: Wiley, 2009, pp. 39–92.

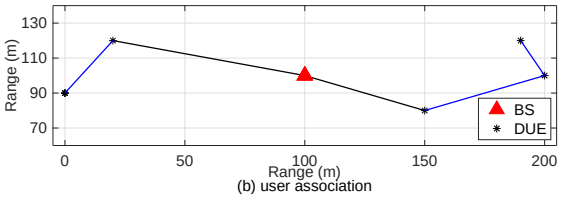


Fig. 2(b). User association from the archived allocation figure; continuation of Fig. 2. This preserves the original evidence, without new experiments.